



Unit 4 · Outcome 1 · Climate Change

Climate change projections: comparing what we saw with what the model says

KK09

VCE Environmental Science · Units 3 & 4 · Exam study summary

Digital Vector EnviroSci · single-topic mastery summary — clean explanations, comparison tables, visual aids, exam traps, command words and worked answers.



Orientation

Where this dot point sits, and exactly what the examiner can ask you to do with it.

"climate change projections, including comparing observed and simulated climate, and the rating of the level of confidence (very high to very low) using the Intergovernmental Panel on Climate Change (IPCC) guidelines."

Strip that sentence down and there are two examinable skills hiding inside it. First, you must be able to compare observed and simulated climate — that is, hold the real measured record next to what a climate model produces, and say how well they match and what that tells us. Second, you must be able to rate confidence in a climate statement the way the IPCC does, on a scale from very high to very low, and justify that rating.

Everything else in this module — scenarios, hindcasting, the projection-vs-prediction distinction — exists to support those two skills. Keep coming back to them.

At Cape Wirreanda. Our running observatory on Victoria's Southern Ocean coast has a 70-year temperature record, tide-gauge data and a feed from the national climate model. Throughout this module the model team is doing two jobs: checking the model against Cape Wirreanda's past, and using it to project the station's future under different emission scenarios.

The mental model

One analogy that makes the whole dot point click.

◆ ANALOGY

Think of a cricket selector's form guide. A selector never says "this batter will score 80 on Saturday." That would be a prediction, and pitches, weather and bowling change everything. Instead they reason in projections: "if the wicket is dry and they bat at three, then a half-century is likely." The "if" is a scenario; the "then" is the projection. And before they trust the form guide at all, they check it against the batter's actual past innings — that's comparing observed with simulated. How sure they are depends on how much data they have and whether the analysts agree — that's confidence.

Hold those four ideas together — observed vs simulated, scenario, projection, confidence — and the rest of this module is just filling in the detail.

Observed vs simulated climate

The centrepiece skill: laying the real record beside the model output. Drag to orbit; toggle the two model runs.

Observed climate is what we actually measured — thermometers, ice cores, tide gauges, satellites. Simulated climate is what a computer climate model (a General Circulation Model) produces when we feed it the physics of the atmosphere, oceans and land, plus a set of forcings (the things that push the energy balance: solar output, volcanoes, and greenhouse gases).

The decisive experiment is to run the model twice over the historical period: once with natural forcings only (sun + volcanoes), and once with natural + human forcings (adding our greenhouse-gas emissions). Then you lay both simulated lines over the observed record and see which one fits.

Notice what happens: the natural-only run wanders but stays roughly flat — the sun and volcanoes alone cannot explain the modern rise. Only when human forcings are added does the simulated line climb to track the observed record. That single picture is the backbone of nearly every "compare observed and simulated" exam answer.

A comparison answer needs a verdict. Weak: "the lines are different." Strong: "the natural-only run fails to reproduce the post-1970 warming, while the natural-plus-human run closely matches the observed record — so the warming is attributable to human forcings."

Hindcasting & attribution

Why running a model backwards is how we earn the right to run it forwards.

Hindcasting (also called a hindcast or backtest) means running the model over a period we already have data for, then checking its output against the observations. If a model can reproduce the twentieth century — the 1991 Pinatubo cooling blip, the warming since the 1970s, the pattern of land warming faster than ocean — we gain confidence that its physics is sound, and therefore that its future projections are credible.

Simulated past closely matches the observed past. The model's equations capture the real climate system, so projections built on the same physics are more believable.

Simulated past drifts from the observed record. Something is missing or mis-tuned; projections from that model carry lower confidence until it's improved.

Attribution is the pay-off of the two-run experiment in §3. Because the model only reproduces observed warming with human forcings included, we can attribute the warming to human activity rather than to natural variability. Comparison isn't just description — it's evidence for cause.

◆ ANALOGY

The diagnostic analogy A doctor trusts a model of your heart only after it reproduces past ECGs. Feeding it "no caffeine" vs "your actual caffeine intake" and finding that only the second run reproduces last week's palpitations is exactly attribution: the symptom is attributable to the caffeine.

Projection vs prediction

A distinction examiners love, because students blur it constantly.

A conditional "if-then" statement of how climate could evolve given a particular scenario of future emissions. Change the scenario and the projection changes. It is not a forecast of any specific year's weather.

An unconditional statement that something will happen — like a weather forecast for Tuesday. Climate science deliberately avoids this for the long-term future because emissions depend on human choices we can't predict.

The reason matters scientifically: the biggest unknown about 2100 isn't the physics of the atmosphere — it's how much greenhouse gas humans will emit. Since that depends on policy, technology and economics, scientists refuse to predict it. Instead they take a scenario as an assumption and project the climate that follows. The honest output is therefore a family of projections, one per scenario.

Strictly, the model projects $\sim 3^{\circ}\text{C}$ under a particular scenario. Drop either the word "projects" or the scenario condition and you've misrepresented the science. In an exam, naming the scenario (e.g. RCP8.5) shows you understand the conditionality.

Scenarios & the projection fan

Slide the emissions pathway and watch the projected warming fan out from a shared past.

The IPCC feeds models a set of standard emission storylines so results are comparable worldwide. The Fifth Assessment (AR5) used Representative Concentration Pathways (RCPs), named for the extra radiative forcing (in watts per square metre) they reach by 2100:

Emissions fall fast, even net-negative late century. Warming held near 1.5–2 °C. The "if we act hard" branch.

Emissions peak then stabilise. Roughly 2–3 °C by 2100. A middle "some action" branch.

Emissions keep rising to mid-century before levelling. Around 3 °C +. Limited action.

Emissions rise throughout the century. Often 4 °C+ — the high-end "business-as-usual" branch.

The Sixth Assessment (AR6) updated these to Shared Socioeconomic Pathways (SSPs) — e.g. SSP1-2.6, SSP5-8.5 — which pair an emissions level with a social/economic storyline. Same idea, richer framing. RCPs remain the standard reference in most VCE materials; recognise both.

Two things to read off the fan. Up to "today" there is one line — the validated, observed past everyone shares. After today it splits: the spread between branches is scenario uncertainty, and it dominates the far-future. Near-term spread, by contrast, comes mostly from natural variability and model differences. That's why scientists are often more confident about 2100 under a given scenario than about any single year along the way.

Rating confidence the IPCC way

The qualitative scale — and the quantitative scale students keep confusing it with.

When the IPCC makes a statement, it tags it with a confidence level from very high to very low. That level is built from two ingredients:

The type, amount, quality and consistency of the evidence — observations, theory, model output, lab and field studies. More and better evidence pushes confidence up.

The degree of agreement among independent studies and experts. When many lines of evidence point the same way, agreement is high.

High evidence + high agreement → very high confidence. Limited evidence + low agreement → very low confidence. Most real statements sit somewhere in between, and the IPCC publishes the rating so readers know how firmly a claim is grounded.

The distinction worth a mark: confidence ≠ likelihood.

Confidence is qualitative — how robust the underlying science is (evidence + agreement). Likelihood is quantitative — a calibrated probability that a specific outcome occurs, with set wording: virtually certain 99–100%, very likely 90–100%, likely 66–100%, about as likely as not 33–66%, unlikely 0–33%, and so on. You only quote a likelihood once confidence is high enough to assign a number at all.

Interactive: the confidence rater

Set the evidence and the agreement, and read off the IPCC confidence — exactly how the matrix works.

Robust evidence and high agreement — e.g. "human activity has warmed the atmosphere, ocean and land."

Try the corners. Robust + high → very high (the headline climate findings). Limited + low → very low (a contested regional detail). The diagonal in between is where most careful science lives.

Command-word decoder

Command word	What the marker wants
Identify / State	Name it — no explanation.
Describe	Give the features / what happens, in order.
Explain	Give the how / why — cause and effect.
Distinguish / Compare	State the difference (and similarity) explicitly.
Analyse	Break into parts and show how they relate.
Evaluate / Justify	Weigh strengths & limits, then a reasoned verdict.

Match your answer to the verb — the fastest marks in the exam.

Exam traps

⚠ EXAM TRAPS

Dropping the scenario condition. Always pair a projected figure with its pathway. Confidence = qualitative (evidence + agreement); likelihood = a calibrated probability. Don't write "very likely confidence." "They're different" earns little. State which run matches the observed record and what that proves. The attribution argument lives in the two-run comparison. Name both forcing sets. "Models are just guesses" misses the point. Say models are validated by reproducing the observed past. If asked to rate confidence, cite the evidence and the agreement that put it there — that's the mark.

P-E-L writing trainer



The P-E-L answer shape — make a Point, support with Evidence, then Link to the question.

🔑 P-E-L WRITING

A reliable structure for the "explain / evaluate" questions on this dot point. Tap each step.

Worked answer: weak vs strong

Same question, two responses. Read what lifts the strong one.

Scientists ran a climate model for 1900–2020 twice — once with natural forcings only, once with natural and human forcings — and compared both with the observed temperature record. Explain how this comparison supports the conclusion that recent warming is human-caused, and comment on the confidence of that conclusion.

The two model runs were different from each other. The one with humans was higher. This shows humans cause warming. Scientists are very likely confident because the model is good.

• **NOTE**

Why it loses marks: no mention of the observed record (the comparison that matters), so attribution isn't actually shown. "Very likely confident" mixes the two scales. "The model is good" isn't a justification.

The natural-only run failed to reproduce the observed warming after ~1970, staying roughly flat, whereas the natural-plus-human run closely matched the observed record. Because only the run including human forcings reproduces what was actually measured, the warming can be attributed to human activity rather than natural variability. Confidence is very high: the evidence is robust (multiple models, long records) and agreement among studies is high.

• **NOTE**

Why it scores: centres the observed record, names both forcing sets, states the attribution explicitly, and rates confidence with the correct ingredients (evidence + agreement).

Practice & self-mark

★ **PRACTICE & SELF-MARK**

Five VCAA-style questions, 21 marks. Write your answer, reveal the model, then mark yourself honestly — your score saves automatically.

EXAM TECHNIQUE

★ **Worked answers — weak vs strong**

The same question answered two ways. The strong column shows the moves that earn the marks.



The P-E-L answer shape — make a Point, support with Evidence, then Link to the question.

✗ WEAK ANSWER

The two model runs were different from each other. The one with humans was higher. This shows humans cause warming. Scientists are very likely confident because the model is good. Why it loses marks: no mention of the observed record (the comparison that matters), so attribution isn't actually shown. "Very likely confident" mixes the two scales. "The model is good" isn't a justification.

✓ STRONG / MODEL ANSWER

The natural-only run failed to reproduce the observed warming after ~1970, staying roughly flat, whereas the natural-plus-human run closely matched the observed record. Because only the run including human forcings reproduces what was actually measured, the warming can be attributed to human activity rather than natural variability. Confidence is very high: the evidence is robust (multiple models, long records) and agreement among studies is high. Why it scores: centres the observed record, names both forcing sets, states the attribution explicitly, and rates confidence with the correct ingredients (evidence + agreement).